

Solution		Marks	Remarks
1. (a)	Amount of energy required $E = mc\Delta T$	1M	
	$= 6 \times 4200 \times (50 - 15)$	1M	
	$= 882000 \text{ J (or 882 kJ)}$	1A	
	Power $P = \frac{E}{t} = \frac{882000}{60}$	3	
	$= 14700 \text{ W (or 14.7 kW)}$		
(b)	Let m kg per minute be the water flow rate	1M	
	$mc\Delta T = Pt$		Or $\frac{m}{t} \propto \frac{1}{\Delta T}$
	$m(4200)(40 - 15) = 14700 \times 60$	1A	$\therefore \frac{m}{6} = \frac{50 - 15}{40 - 15}$
	$m = 8.4 \text{ (kg min}^{-1} \text{ or kg)}$	2	
2. (a)	$n = \frac{pV}{RT} \propto pV \text{ (for constant } T)$	1M	
	$\frac{n_A}{n_B} = \frac{(p)(3V)}{(2p)(2V)}$		
	$n_A = 0.75 \times 0.8 \text{ mol}$		
	$= 0.6 \text{ (mol)}$	1A	

Or $(2p)(2V) = 0.8RT$		1M
$pV = 0.2RT$		
$p(3V) = nRT$		1A
$n = 3 \times 0.2 = 0.6 \text{ (mol)}$		
		2
(b) (i)	$n = n_A + n_B$	1M
	$p'(3V + 2V) = p(3V) + (2p)(2V)$	1A
	$p' = 1.4p$	
Or $p'(3V + 2V) = (0.6 + 0.8)RT$		1M
$p'(5V) = 1.4RT$		1A
$p' = 1.4p$		
		2
(ii)	When the tap is open, number of gas molecules in vessel A increases due to the (net) flow of molecules from B to A, the frequency of collision of gas molecules with the vessel's wall increases, pressure increases.	1A
		1A
		2

3.	(a)	If the maximum load is exceeded, the braking distance will increase if the friction provided remains the same.	1A	
		Vehicles would not be able to stop in time in case of emergency (thus dangerous).	1A	
	(b)	<u>Or</u> A larger friction is required in braking the vehicle within the same distance, accident may occur if the brakes cannot provide such frictional forces.	1A	
			1A	
			2	
	(i)	If the brakes are applied continuously, thermal energy generated will heat up the brake pads / brakes to too high a temperature that the brakes may fail.	1A	
			1	
	(ii)	Let D be the distance travelled along the ramp. Kinetic energy of vehicle becomes its gravitational potential energy:	1M	<u>Or</u> work done against vehicle's weight component along the ramp by its kinetic energy: $\frac{1}{2}mv^2 = mg \sin \theta \times D$
		$\frac{1}{2}m(25)^2 = m(9.81)(D \sin 30^\circ)$		
		$D = 63.710499 \text{ m}$		
		$\approx 63.7 \text{ m (62.5 m for } g = 10 \text{ m s}^{-2}\text{)}$	1A	
		<u>Or</u> $v^2 = u^2 + 2as$	1M	mass of vehicle = m Note: $D \sin 30^\circ = 31.8552 \text{ m}$

$v = 25 + 2(-9.81 \sin 50^\circ) D$ $D \approx 63.7 \text{ m}$		1A	Accept $D = 62.0 \text{ m to } 64.0 \text{ m}$
		2	
4. (a) (i)	$\text{K.E.} + \text{P.E.} = \frac{1}{2}(0.3)(4)^2 + (0.3)(9.81)(0.2)$ $= 2.4 + 0.5586 = 2.9886 \text{ J}$ $\approx 2.99 \text{ J (3.0 J for } g = 10 \text{ m s}^{-2})$	1M+1M 1A	$= 2.4 + 0.6 = 3.0 \text{ J for } g = 10 \text{ m s}^{-2}$
		3	
(ii)	As the spring gun is fixed, there is external force acting on the system / the gun, total momentum (of the spring gun and cannon ball) is not conserved.	1M 1A	
		2	
(b)	Vertical : $s = ut + \frac{1}{2}at^2$ $0 = (4 \sin 50^\circ) t_f - \frac{1}{2}(9.81) t_f^2$ $t_f = 0.624705 \text{ s (0.612836 s for } g = 10 \text{ m s}^{-2})$ $\approx 0.625 \text{ s (0.613 s for } g = 10 \text{ m s}^{-2})$ Horizontal : $R = 4 \cos 50^\circ \times t_f = 4 \cos 50^\circ \times 0.625$ $= 1.606210 \text{ m}$ $\approx 1.61 \text{ m (1.57 m for } g = 10 \text{ m s}^{-2})$	1M 1A 1M 1A	Or $\frac{t_f}{2} = \frac{u \sin \theta}{g} = \frac{4 \sin 50^\circ}{9.81}$ Accept $t_f = 0.61 \text{ s to } 0.63 \text{ s}$ Or $R = u \cos \theta \times \frac{2(u \sin \theta)}{g}$ Accept $R = 1.56 \text{ m to } 1.62 \text{ m}$
		4	
(c)	t_f increases since the initial vertical velocity / component is greater.	1A 1A	
		2	

Solution		Marks	Remarks				
5. (a)	(i) m (5.0 cm) = 50 g (10.0 cm) $m = 100$ g or 0.1 kg	1M 1A					
		2					
	(ii) Counter-weight position : 10.0 cm $\pm$ 0.1 cm Percentage error = $100\% \times \left(\frac{0.1}{10.0}\right) = 1\%$ $\therefore m = 101$ g to 99 g i.e. maximum error = ± 1 g	1M 1A					
		2					
	(b) Spring balance reading = $mg = (0.1 \text{ kg}) (9.81 \text{ N kg}^{-1})$ $= 0.981 \text{ N}$ (1.0 N for $g = 10 \text{ N kg}^{-1}$) $\approx 1.0 \text{ N}$	1M/1A					
		1					
	(c) (i)						
	<table><tr><td>counter-weight position on beam balance</td><td>spring balance reading</td></tr><tr><td>the same</td><td>reading increases</td></tr></table>	counter-weight position on beam balance		spring balance reading	the same	reading increases	1A+1A
	counter-weight position on beam balance	spring balance reading					
	the same	reading increases					
	2						
(ii) The beam balance would fail to work / to measure the mass of the load, as the apparent weight is zero (or weightless), the counter-weight can take any position.	1A 1A						
	2						

6. (a) Place the (cylindrical) lens on the piece of paper and trace its outline, use ray tracing method e.g.
 - direct and trace a light ray towards the lens.
 - direct and trace a light ray along / parallel to the principal axis
- Direct another light ray parallel to the first one / principal axis (by shifting the ray box) and trace the path(s) of the (emerging) ray(s) on the paper.
 Extend (the path of) the emerging rays (backward) and locate the point of intersection (on the focal plane containing F).
 Measure the distance from the point of intersection (or F) to the centre of the lens, which gives the focal length of the lens.
- Source of error:
 Scale uncertainty/accuracy/precision of the plastic ruler (read to nearest mm).
- OR Unable to mark the path correctly because of the thickness of the beam of light from the ray box
 OR The ray(s) is/are not parallel (to the principal axis)
 OR Any reasonable answer

1A

Or Using lens formula:

- use ray tracing method: direct and trace a light ray (not parallel to the principal axis) towards the lens

1A

- extend (the path of) the emerging ray backward and locate the point of intersection (with the principal axis).

1A

- measure the corresponding object distance u and image distance v (on the principal axis)

1A

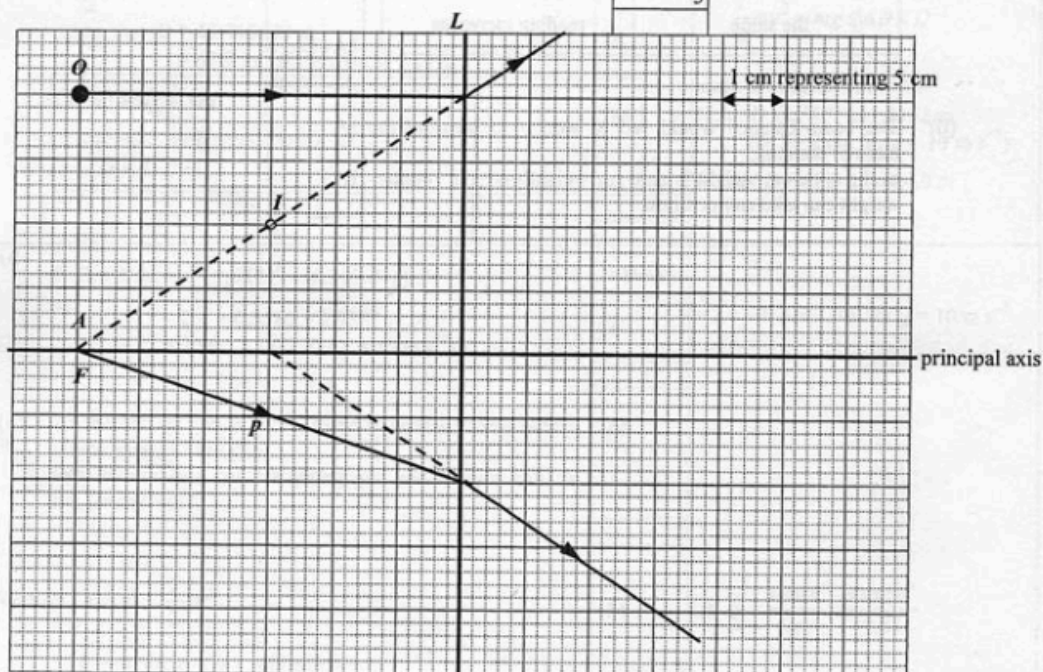
- sub. u and v into lens formula to find f

1A

Accept: Using diagram(s) to supplement / simplify the description.

5

(b)



- (i) L is diverging/concave.
 - only a diverging lens can produce a (virtual / erect / diminished) image between the object and the lens

1A

1A

2

- (ii) Focal length = 30 cm
 Correct ray to find F

1A

1A

Accept (30 ± 1) cm

2

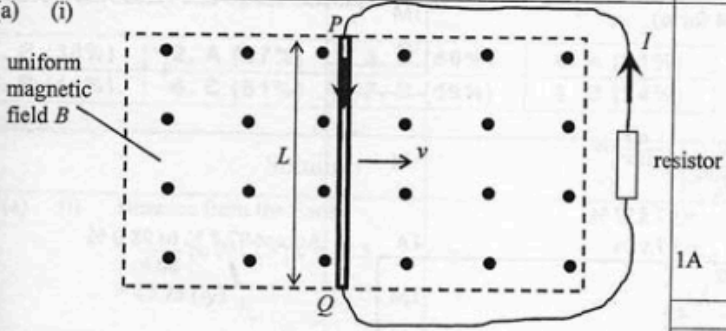
- (iii) Correct ray p

1A

1

Solution		Marks	Remarks
7. (a)	(i) Increase the separation between the double slit and the screen, D .	1A	
		1	
	(ii) The separation of the bright dots on the screen becomes larger, thus the percentage error in its measurement is smaller.	1A	
		1	
	(iii) The angular position of the 2 nd order bright dot $\theta = \tan^{-1}\left(\frac{1.56/2}{1.40}\right) = 29.124053^\circ$	1M	
	Grating spacing $d = \frac{10^{-3}}{400} = 2.5 \times 10^{-6} \text{ m}$	1M	
	Applying $d \sin \theta = n\lambda$, Wavelength $\lambda = \frac{2.5 \times 10^{-6} \times \sin 29.12^\circ}{2}$ $= 6.08378 \times 10^{-7} \text{ m}$ $\approx 6.08 \times 10^{-7} \text{ m} (= 608 \text{ nm})$	1A	Accept $\lambda = (6.06 \text{ to } 6.10) \times 10^{-7} \text{ m}$
		3	
	(b) (i) The equation can only be applied for - $\lambda \ll a$ (i.e. wavelength $\ll$ separation of the two sources), OR λ is much smaller than a - $a \ll D$ (i.e. separation of the two sources $\ll$ separation of the sources and detector), OR a is much smaller than D	Any ONE 1A	Note : the order of magnitude of the wavelength of sound is about 10^{-1} m
	Or Using the fringe separation equation to find the wavelength of sound is not accurate for - λ is comparable to/ NOT much smaller than a - a is NOT much smaller than D	Any ONE 1A	
(ii)	For the 1 st order maximum, wavelength $\lambda = \text{path difference } PB - PA$ $= \sqrt{(1+0.5)^2 + 2^2} - \sqrt{(1-0.5)^2 + 2^2}$ $= 2.5 - 2.06155281 = 0.43844719 \text{ m} \approx 0.438 \text{ m}$	1M	
	speed of sound: $v = f\lambda = 750 \times 0.4384$ $= 328.835 \text{ m s}^{-1}$ $\approx 329 \text{ m s}^{-1}$	1M	
		1A	Accept $v = 328 \text{ m s}^{-1}$ to 330 m s^{-1}
		3	

8. (a) (i) To terminals X	1A	
	1	
(ii) $P = IV$ $800 \text{ W} = I(220 \text{ V})$ $I = 3.636364 \text{ A}$ $\approx 3.64 \text{ A}$	1M 1A	
	2	
(iii) $800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$ $P_{\text{keep warm}} = \frac{V^2}{4R}$ $= 800\left(\frac{1}{5}\right) = 160 \text{ W}$	1M 1M 1A	
<u>Alternative (I)</u> $800 = \frac{V^2}{R} + \frac{V^2}{4R} = \frac{5V^2}{4R}$ $R = 75.625 \Omega$ $P_{\text{keep warm}} = \frac{V^2}{4R}$ $= \frac{220^2}{4(75.625)}$ $= 160 \text{ W}$	1M 1M 1A	
<u>Alternative (II)</u> Power $\propto \frac{1}{\text{resistance}}$ $\frac{P_{\text{keep warm}}}{P_{\text{heating}}} = \frac{R_{\text{eq}}}{4R}$ $P_{\text{keep warm}} = P_{\text{heating}} \left(\left(\frac{1}{R} + \frac{1}{4R} \right)^{-1} / 4R \right)$ $= 160 \text{ W}$	1M 1M 1A	
	3	
(b) (i) Electrical energy (consumed)	1A	
	1	
(ii) (1) only the fuse blows	1A	
(2) only the RCCB cuts off the supply	1A	
	2	

Solution	Marks	Remarks
9. (a) (i) 	1A 1	
(ii) By Lenz's law, a magnetic force F_B acts on the rod such that it opposes the rod's motion, an external force F is needed to balance F_B so as to maintain uniform motion (or constant v) $\therefore F = F_B = ILB$ (in magnitude) Or $F \cdot v = I \xi, \therefore F = \frac{I \xi}{v}$	1A 1A 1A 1A 3	Accept : Work done by an external force is needed to transfer mechanical energy into electrical energy.
(iii) mechanical power input $= Fv$ $\quad \quad \quad = (ILB) v$ power input = power output (electrical) $ILB v = I \xi$ $\xi = BLv$	1M 1M 2	
(b) (i) Horizontal (component) is perpendicular to the mast. Or Vertical (component) is parallel to the mast.	1A 1A 1	
(ii) $\xi = (B \cos 30^\circ) L v$ $\quad = (50 \times 10^{-6} \cos 30^\circ) (20) (6)$ $\quad = 5.196152 \times 10^{-3} \text{ V}$ $\quad \approx 5.20 \text{ mV}$ more electrons at end X	1M 1A 1A 3	by (a)(iii) using the horizontal component of B
(iii) No current. Both the cable and the mast cut the field lines in the same way, the e.m.f.'s produced are equal and thus oppose each other. Or No change in magnetic flux through the loop of the circuit.	1A 1A 1A 2	

10. (a)	(i)	$226 - 206 = 20$ (multiple of 4 for α) $\therefore {}^{206}_{82}\text{Pb}$ is the end product	1M	Accept 97.8 % to 98.0 %
			1A	
	(ii)	% undecayed Ra-226 left $\frac{N}{N_0} = e^{\frac{\ln 2}{1600} \times 50}$ $= 97.857\%$ $\approx 97.9\%$	2	
			1M	
			1A	
			1M	
			1A	
			2	
	(b)	(i)	1A	
			1	
		(ii)	1A	
			1	
		(iii)	1A	
			1A	